1.

Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

The diagram above shows the main hardware components of a desktop computer: Motherboard, CPU, RAM, GPU, HDD, and SSD.

Functions of Each Component

1. Motherboard (Main Circuit Board): Connects all computer components and allows them to communicate.

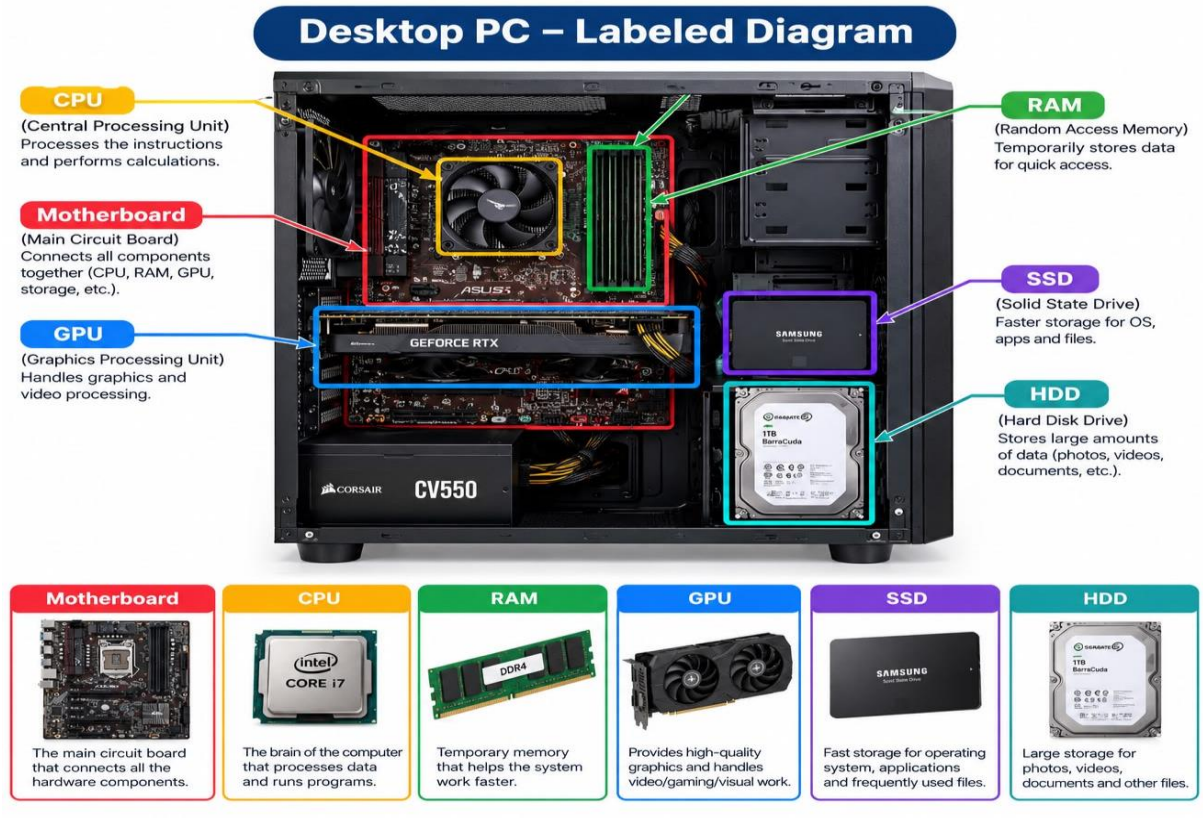
2. CPU (Central Processing Unit): Processes instructions and performs calculations. It is known as the brain of the computer.

3. RAM (Random Access Memory): Temporarily stores data and instructions currently being used by the computer.

4. GPU (Graphics Processing Unit): Processes graphics, images, videos, and visual effects.

5. HDD (Hard Disk Drive): Stores large amounts of data, including documents, photos, videos, and software.

6. SSD (Solid State Drive): Provides fast storage and helps the computer boot and load applications qu



2.Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

2. Comparison Between HDD and SSD

HDD (Hard Disk Drive) and **SSD (Solid State Drive)** are storage devices used to store data, files, software, and operating systems in a computer.

Comparison Table: HDD vs SSD

Sr. No.	Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
1	**Speed**	Slower in reading and writing data.	Faster in reading and writing data.
2	**Technology**	Uses spinning magnetic disks and a moving read/write head.	Uses flash memory and has no moving parts.
3	**Durability**	Less shock-resistant because it contains moving parts.	More shock-resistant because it has no moving parts.
4	**Boot Time**	Takes more time to start the operating system.	Starts the operating system quickly.
5	**Noise**	May produce noise due to moving parts.	Works silently.
6	**Power Consumption**	Generally consumes more power.	Generally consumes less power.
7	**Cost**	Cheaper per GB of storage.	More expensive per GB of storage.
8	**Storage Capacity**	Available in large capacities at a lower price.	Available in different capacities, but usually costs more for the same storage size.
9	**Typical Use Cases**	Used for storing large files, videos, photos, backups, and archives.	Used in laptops, gaming PCs, and computers that require fast performance.
10	**Performance**	Slower application loading and file transfers.	Faster application loading, file transfers, and gaming load times.

3.

Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

Role of CPU, RAM, and GPU in Running a Game

1. CPU (Central Processing Unit):
The CPU is the brain of the computer. It processes game instructions, manages player movements, controls game logic, and handles other tasks while playing games like PUBG or Free Fire on a PC.

2. RAM (Random Access Memory):
RAM temporarily stores the data and information required by the game while it is running.

More RAM helps the computer run the game smoothly, reduces lag, and allows other applications to run in the background.

3. GPU (Graphics Processing Unit):
The GPU is responsible for rendering graphics, textures, animations, and visual effects in the game. A powerful GPU provides better graphics, smoother gameplay, and higher frame rates in games like PUBG or Free Fire.

4.
Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Hardware Components to Upgrade a PC for Running the Latest FIFA Game Smoothly

To run the latest version of FIFA (EA SPORTS FC) smoothly on a PC, I would consider upgrading the following hardware components:

1. CPU (Central Processing Unit):
I would upgrade the CPU to a faster and more powerful processor. A good CPU helps process game instructions, player movements, and game logic efficiently, improving overall performance.

2. GPU (Graphics Processing Unit):
I would upgrade the GPU to a powerful graphics card. It helps render high-quality graphics, realistic player animations, and smooth gameplay with better frame rates.

3. RAM (Random Access Memory):
I would increase the RAM to at least 16 GB, provided the game's system requirements support it. Sufficient RAM helps the game run smoothly and reduces performance issues when multiple applications are open.

4. Storage (SSD):
I would install the game on an SSD instead of an HDD. An SSD provides faster loading times, helps the game start quickly, and reduces waiting time when loading matches.

5. Motherboard:
I would upgrade the motherboard if necessary to support the new CPU, RAM, or GPU. It must be compatible with the upgraded components to ensure proper functioning.

6. Power Supply Unit (PSU):
I would upgrade the PSU if the new CPU or GPU requires more power. A reliable power supply provides stable electricity to all components and helps prevent unexpected shutdowns.